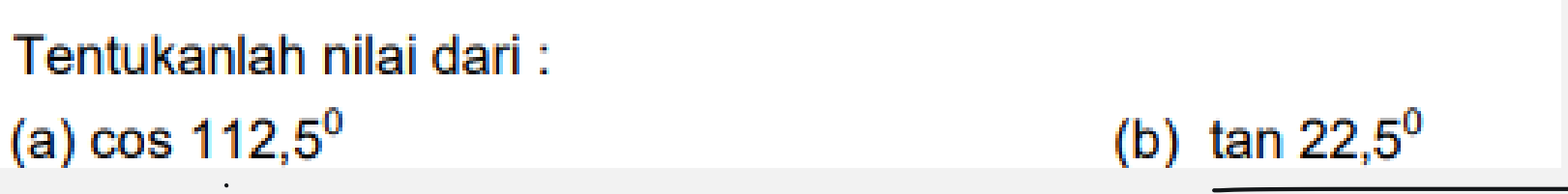
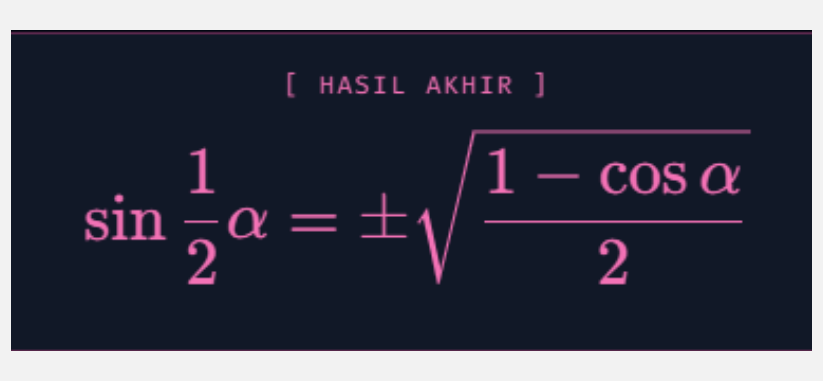
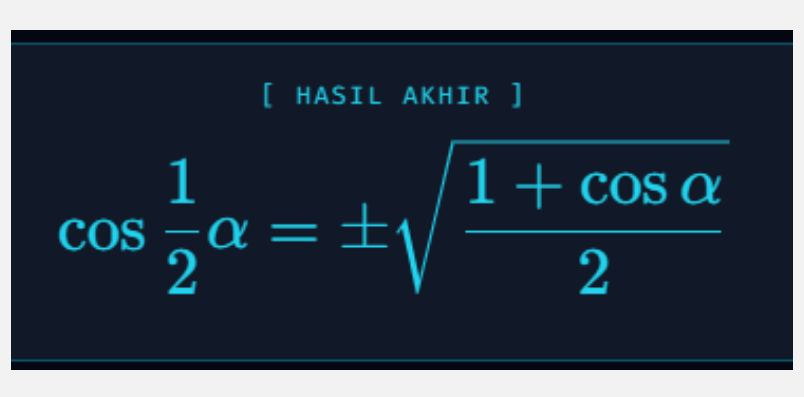

Tentukanlah nilai dari :

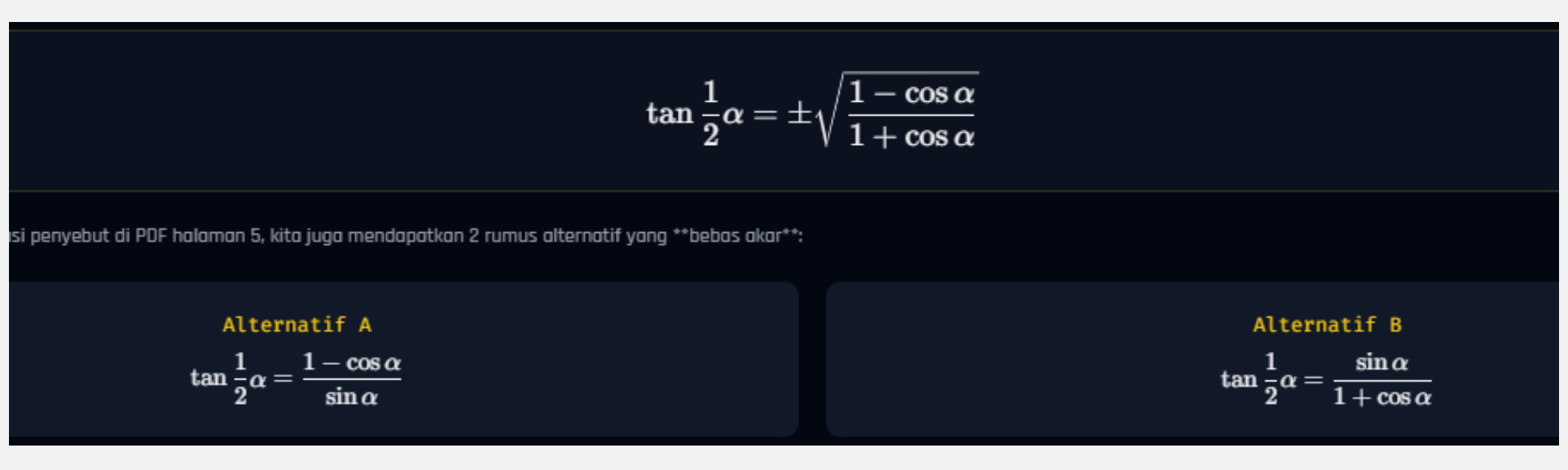
(a) cos 112,5⁰ (b) tan 22,5⁰


[HASIL AKHIR]

$$\sin\frac{1}{2}\alpha = \pm\sqrt{\frac{1-\cos\alpha}{2}}$$


[HASIL AKHIR]

$$\cos\frac{1}{2}\alpha = \pm\sqrt{\frac{1+\cos\alpha}{2}}$$


$$\tan\frac{1}{2}\alpha = \pm\sqrt{\frac{1-\cos\alpha}{1+\cos\alpha}}$$

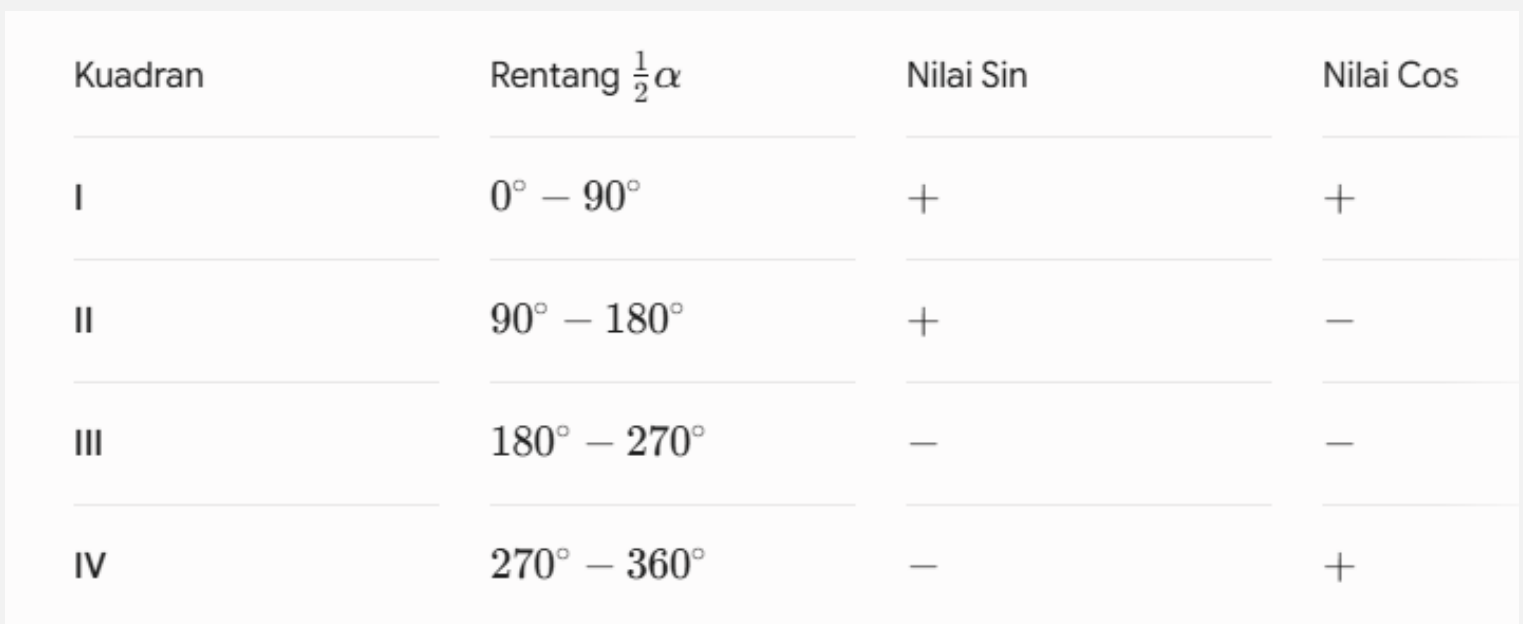
si penyebut di PDF halaman 5, kita juga mendapatkan 2 rumus alternatif yang **bebas akar**:

Alternatif A

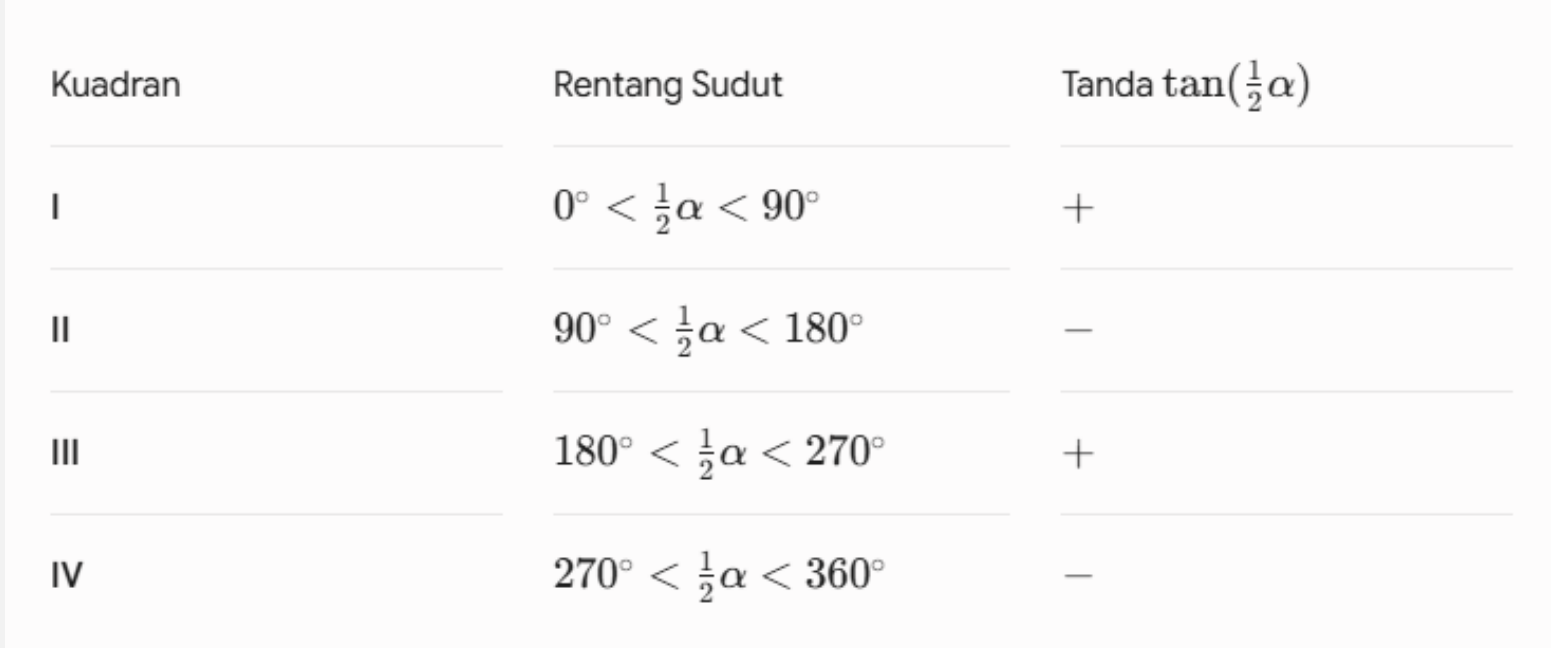
$$\tan\frac{1}{2}\alpha = \frac{1-\cos\alpha}{\sin\alpha}$$

Alternatif B

$$\tan\frac{1}{2}\alpha = \frac{\sin\alpha}{1+\cos\alpha}$$



Kuadran	Rentang $\frac{1}{2}\alpha$	Nilai Sin	Nilai Cos
I	$0^\circ - 90^\circ$	$+$	$+$
II	$90^\circ - 180^\circ$	$+$	$-$
III	$180^\circ - 270^\circ$	$-$	$-$
IV	$270^\circ - 360^\circ$	$-$	$+$



Kuadran	Rentang Sudut	Tanda $\tan(\frac{1}{2}\alpha)$
I	$0^\circ < \frac{1}{2}\alpha < 90^\circ$	$+$
II	$90^\circ < \frac{1}{2}\alpha < 180^\circ$	$-$
III	$180^\circ < \frac{1}{2}\alpha < 270^\circ$	$+$
IV	$270^\circ < \frac{1}{2}\alpha < 360^\circ$	$-$

$\cos\frac{1}{2}(225) = \pm\sqrt{\frac{1+\cos(225)}{2}}$

$\cos\frac{1}{2}(225) = -\sqrt{\frac{1-\frac{1}{2}\sqrt{2}}{2}}$

$\cos\frac{1}{2}(225) = -\sqrt{\frac{\frac{2}{2}-\frac{1}{2}\sqrt{2}}{2}}$

$\cos\frac{1}{2}(225) = -\sqrt{\frac{\frac{2-\sqrt{2}}{2}}{2}}$

$= -\sqrt{\frac{2-\sqrt{2}}{2} \times \frac{1}{2}}$

$= -\sqrt{\frac{2-\sqrt{2}}{4}}$ ✓

$= -\frac{1}{2}\sqrt{2-\sqrt{2}}$

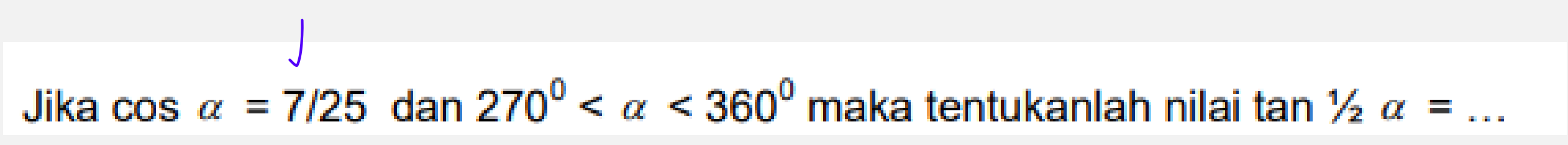
$\tan\frac{1}{2}(45) = \frac{\sin 45^\circ}{1+\cos 45}$

$= \frac{\frac{1}{2}\sqrt{2}}{1+\frac{1}{2}\sqrt{2}}$

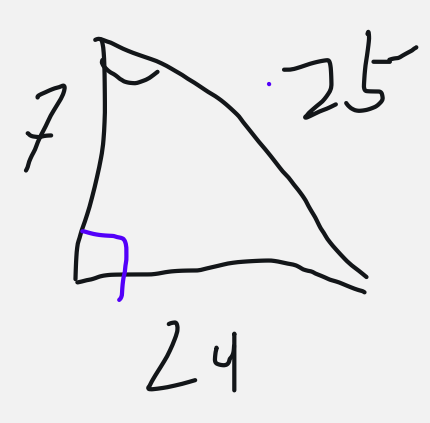
$= \frac{\frac{\sqrt{2}}{2}}{\frac{2+\sqrt{2}}{2}}$

$= \frac{\sqrt{2}}{\cancel{2}} \times \frac{\cancel{2}}{2+\sqrt{2}}$

$= \frac{\sqrt{2}}{2+\sqrt{2}} \times \frac{2-\sqrt{2}}{2-\sqrt{2}} = \frac{2\sqrt{2}-2}{4-2} = \frac{2\sqrt{2}-2}{2} = \sqrt{2}-1$


✓

Jika cos α = 7/25 dan 270⁰ < α < 360⁰ maka tentukanlah nilai tan ½ α = …



$\sin\alpha = -\frac{24}{25}$

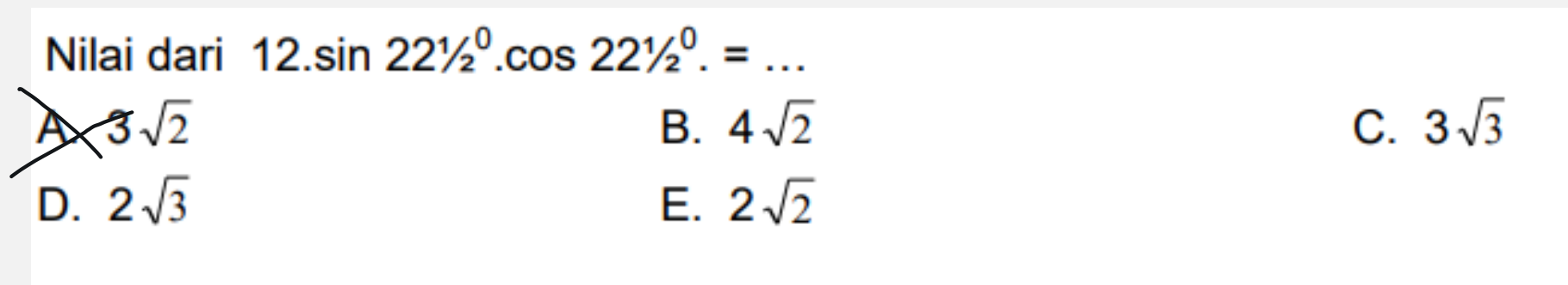
$\tan\frac{1}{2}\alpha = \frac{-\frac{24}{25}}{1+\frac{7}{25}}$

$= \frac{-\frac{24}{25}}{\frac{25}{25}+\frac{7}{25}}$

$= \frac{-\frac{24}{\cancel{25}}}{\frac{32}{\cancel{25}}}$

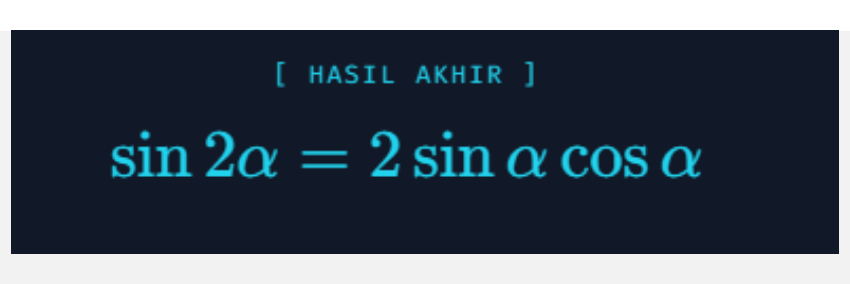
$= -\frac{24}{32}$

$= -\frac{3}{4}$ ✓


Nilai dari 12.sin 22½⁰.cos 22½⁰. = …

~~A.~~ $3\sqrt{2}$ B. $4\sqrt{2}$ C. $3\sqrt{3}$

D. $2\sqrt{3}$ E. $2\sqrt{2}$


[HASIL AKHIR]

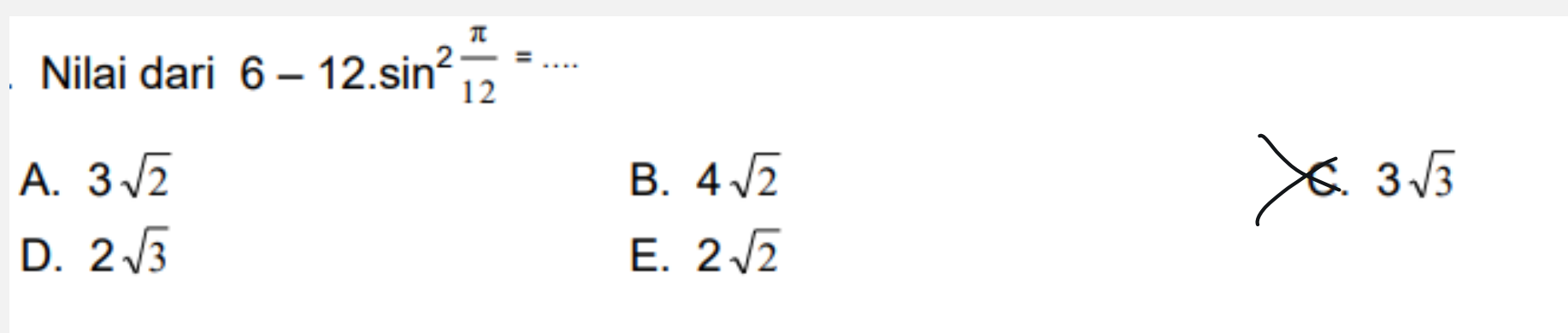
$$\sin 2\alpha = 2\sin\alpha\cos\alpha$$

$= 6.2.\sin 22{,}5 . \cos 22{,}5$

$= 6\sin 2(22{,}5)$

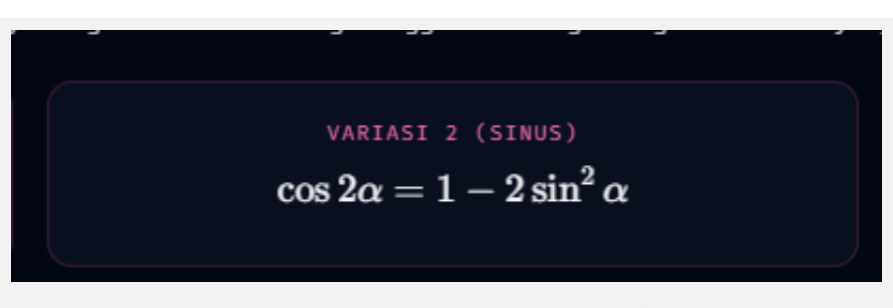
$= 6 . \frac{1}{2}\sqrt{2}$

$= 3\sqrt{2}$


Nilai dari 6 – 12.sin² $\frac{\pi}{12}$ = ...

A. $3\sqrt{2}$ B. $4\sqrt{2}$ ~~C.~~ $3\sqrt{3}$

D. $2\sqrt{3}$ E. $2\sqrt{2}$


VARIASI 2 (SINUS)

$$\cos 2\alpha = 1 - 2\sin^2\alpha$$

$= 6(1-2.\sin^2\frac{\pi}{12})$

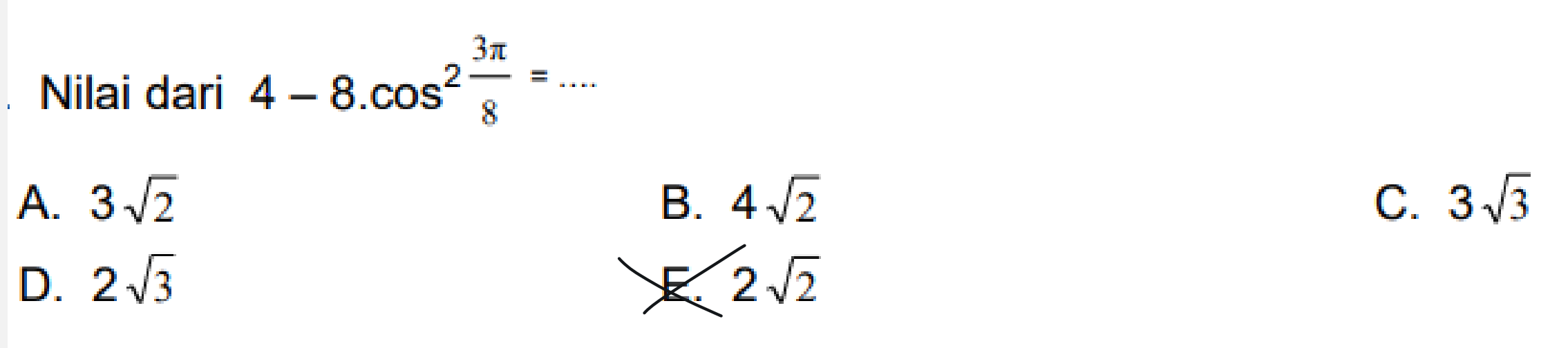
$= 6(\cos 2\frac{\pi}{\cancel{12}6})$

$= 6(\cos\frac{180}{6})$

$= 6(\cos 30)$

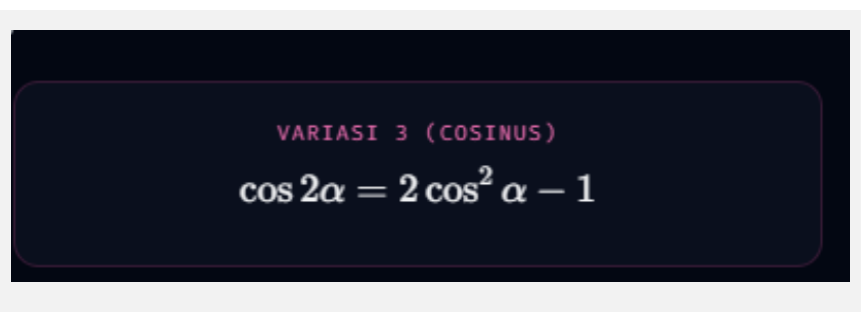
$= 6(\frac{1}{2}\sqrt{3})$

$= 3\sqrt{3}$


Nilai dari 4 – 8.cos² $\frac{3\pi}{8}$ = ...

A. $3\sqrt{2}$ B. $4\sqrt{2}$ C. $3\sqrt{3}$

D. $2\sqrt{3}$ ~~E.~~ $2\sqrt{2}$


VARIASI 3 (COSINUS)

$$\cos 2\alpha = 2\cos^2\alpha - 1$$

$= 4(1-2.\cos^2\frac{3\pi}{8})$

$= -4(-1+2.\cos^2\frac{3\pi}{8})$

$= -4(2.\cos^2\frac{3\pi}{8}-1)$

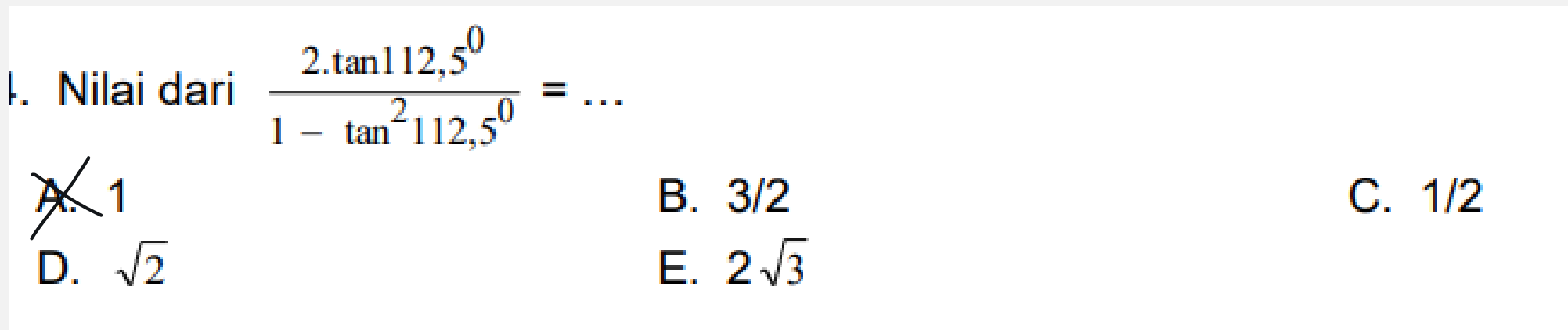
$= -4(\cos 2\frac{3\pi}{\cancel{8}4})$

$= -4(\cos\frac{54\cancel{0}}{4})$

$= -4(\cos 135)$

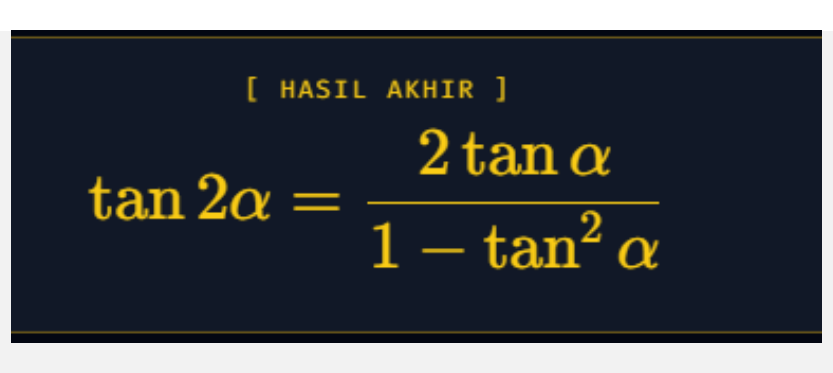
$= -\cancel{4}(-\frac{1}{\cancel{2}}\sqrt{2})$

$= 2\sqrt{2}$


Nilai dari $\frac{2.\tan 112{,}5^0}{1-\tan^2 112{,}5^0}$ = …

~~A.~~ 1 B. 3/2 C. 1/2

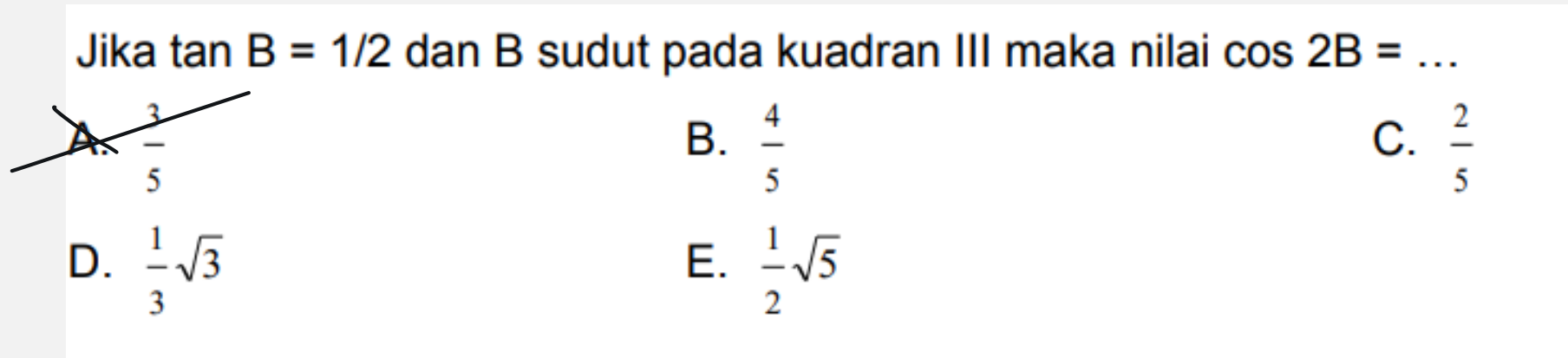
D. $\sqrt{2}$ E. $2\sqrt{3}$


[HASIL AKHIR]

$$\tan 2\alpha = \frac{2\tan\alpha}{1-\tan^2\alpha}$$

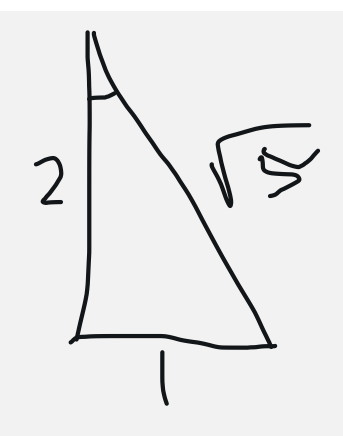
$\tan 2(112{,}5) = \tan 225$

$= 1$


Jika tan B = 1/2 dan B sudut pada kuadran III maka nilai cos 2B = …

~~A.~~ $\frac{3}{5}$ B. $\frac{4}{5}$ C. $\frac{2}{5}$

D. $\frac{1}{3}\sqrt{3}$ E. $\frac{1}{2}\sqrt{5}$



$\cos 2B = 2\cos^2 B - 1$

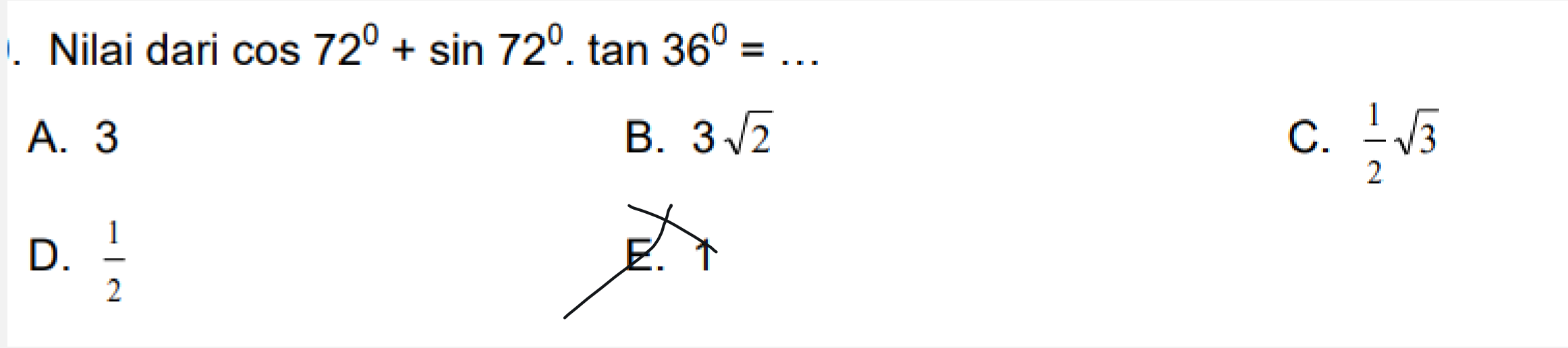
$\cos 2B = 2(-\frac{2}{\sqrt{5}})^2 - 1$

$= 2 . \frac{4}{5} - 1$

$= \frac{8}{5} - 1$

$= \frac{8}{5} - \frac{5}{5}$

$= \frac{3}{5}$


Nilai dari cos 72⁰ + sin 72⁰. tan 36⁰ = …

A. 3 B. $3\sqrt{2}$ C. $\frac{1}{2}\sqrt{3}$

D. $\frac{1}{2}$ ~~E.~~ 1

✓

$\cos 2(36) + \sin 2(36) . \tan(36)$

$1 - 2\sin^2(36) + 2\sin(36)\cancel{\cos(36)} . \frac{\sin(36)}{\cancel{\cos(36)}}$

$1 - 2\sin^2(36) + 2\sin^2(36)$

1